

1. In a very quiet place and in a resting condition, tip-link extends 5 nm at a resting force of 8 pN and frequency of 20 Hz. Calculate the power consumption of each tip-link in a resting state. (3)
2. For a fluorescence microscope using 500 nm excitation light and an objective with a numerical aperture (NA) of 1.6, calculate Abbe's lateral resolution limit. (2)
3. Fluorescence Intensity profile of a single GFP (a point source) emitting light onto a CCD camera is shown below. This is known as Airy pattern. (a) Draw the corresponding Intensity Profile along the white line shown on the Figure. (b). Green line marks 440 nm. If the GFP emitting at a peak of 510 nm, and the numerical aperture of the air objective is 1.4, find out the maximum light capture angle of the objective for this image. (2 + 3 = 5)



4. Based on the force-response of biological bonds, they are of three types: (a) Define all three types of bonds with figures. (b) Motor protein exerts forces that are typically lower than 10 pN. Which of the force-estimation techniques will be most suitable for tracking the motor protein? (3+2 = 5)

$$d = \frac{s}{t}$$

$$f = \frac{1}{t} = \frac{1}{20}$$

$$\eta \sim R.I.$$